SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**JANI DILIP**
Register Number	**192424314**

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for realworld applications.* (BL3)

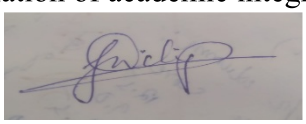
Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**

Question Count: 3

Code of Conduct

I **Jani Dilip (192424314)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks - Case Study 1. **Case Study 1: Smart Retail Inventory Management System**

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.
Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Wellstructured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

A retail chain uses a cloud-based inventory management system to manage products, monitor stock levels, and analyze sales across multiple stores. Store devices such as Point of Sale (POS) systems and dashboards communicate with centralized cloud services through APIs. The cloud platform stores inventory data, transaction records, and analytics datasets while providing demand forecasting and reporting services.

Key Components and Working

- Clients: POS systems, barcode scanners, inventory dashboards, and mobile devices are used by employees to record sales, check stock, and generate reports.
- Web APIs: APIs connect the client applications with cloud services, allowing inventory updates, product searches, and sales transactions in real time.
- Cloud Storage: Stores product details, inventory records, sales transactions, customer information, and analytics data securely with backup and recovery support.
- Platform Services (PaaS): Provide analytics, demand forecasting, report generation, and machine learning capabilities to predict future stock requirements.
- Infrastructure (IaaS): Uses virtual machines, networking, storage, and load balancers to ensure system reliability and scalability.
- Security: VPNs, firewalls, API gateways, authentication, and encryption protect business data and secure communication between stores and cloud services.

Advantages

- Real-time inventory monitoring across all stores.
- Automatic stock updates after every sale.
- Better demand forecasting using analytics.
- Reduced inventory shortages and excess stock.
- Secure and scalable cloud infrastructure.
- Improved operational efficiency and decision-making.

Cloud Service Models Used

- IaaS: Virtual machines, networking, and storage.
- PaaS: Analytics and demand forecasting services.
- SaaS: Inventory management dashboards and applications.

Case Study 2: Cloud-Based Document Collaboration System

A cloud-based document collaboration system allows users to create, edit, and share documents from any location using web browsers or mobile devices. The system supports real-time collaboration, version control, and secure document storage through cloud computing technologies.

Key Components and Working

- Clients: Users access the application through web browsers, tablets, laptops, or mobile devices to create and edit documents.
- Web APIs: APIs synchronize document changes across multiple users and devices, enabling real-time collaboration.
- Distributed Cloud Storage: Stores documents, images, user profiles, and previous versions across multiple servers to ensure high availability and fault tolerance.
- Version Control: Maintains document history, allowing users to restore previous versions whenever required.
- Network Infrastructure: Uses distributed servers, load balancing, and content delivery mechanisms to provide low latency and high availability worldwide.
- Security: Encryption, identity management, authentication, and access permissions ensure that only authorized users can view or edit documents.

Advantages

- Real-time document editing by multiple users.
- Automatic synchronization across devices.
- Secure document sharing and collaboration.
- Backup and recovery of previous document versions.
- High availability and reliable cloud storage.
- Easy access from anywhere with an internet connection.

Cloud Service Models Used

- SaaS: Online document editing application.
- PaaS: Collaboration and synchronization services.
- IaaS: Cloud servers, networking, and storage infrastructure.

Case Study 3: Online Food Ordering Platform

An online food ordering platform enables customers to browse restaurant menus, place orders, make payments, and track deliveries through web browsers and mobile applications. The platform uses cloud computing to provide scalable, secure, and reliable services.

Key Components and Working

- Clients: Customers use web browsers or mobile applications to search restaurants, place orders, and monitor deliveries.
- RESTful Web APIs: APIs connect the frontend applications with backend cloud services for user authentication, menu retrieval, order placement, payment processing, and order tracking.
- Cloud Storage: Stores menu images, customer profiles, invoices, payment records, and order history securely.
- Infrastructure (IaaS): Uses virtual machines, networking resources, and storage to host backend services.
- Managed Kubernetes (PaaS): Deploys and manages containers automatically, ensuring high availability, scalability, and efficient resource utilization.
- Security: HTTPS, authentication tokens (JWT), role-based access control (RBAC), and encryption protect customer data and online transactions.

Advantages

- Fast and efficient food ordering process.
- Automatic scaling during peak demand.
- Secure payment and customer information.
- High system availability and reliability.
- Real-time order tracking and notifications.
- Simplified maintenance using cloud services.

Cloud Service Models Used

- IaaS: Virtual machines, networking, and storage.
- PaaS: Kubernetes, backend services, and application management.
- SaaS: Online food ordering application for customers.